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Feature-aware unsupervised learning with joint variational attention and automatic clustering

Ru Wang*, Lin Li†, Peipei Wang†, Xiaohui Tao‡, Peiyu Liu*,§
*School of Information Science and Engineering, Shandong Normal University, Jinan,250358, China
Email: ruwang0929@gmail.com, liupy@sdnu.edu.cn
†School of Computer Science and Technology, Wuhan University of Technology,Wuhan, 430070, China
Email: cathylilin@whut.edu.cn, ppwang07@whut.edu.cn
‡School of Sciences, University of Southern Queensland, Toowoomba, Australia
Email: Xiaohui.Tao@usq.edu.au
§ Corresponding auther

***Abstract*—Deep clustering aims to cluster unlabeled real-world samples by mining deep feature representation. Most of existing methods remain challenging when handling high-dimensional data and simultaneously exploring the complementarity of deep feature representation and clustering. In this paper, we propose a novel Deep Variational Attention Encoder-decoder for Clustering (DVAEC). Our DVAEC improves the representation learning ability by fusing variational attention. Specifically, we design a feature-aware automatic clustering module to mitigate the unreliability of similarity calculation and guide network learning. Besides, to further boost the performance of deep clustering from a global perspective, we define a joint optimization objective to promote feature representation learning and automatic clustering synergistically. Extensive experimental results show the promising performance achieved by our DVAEC on six datasets comparing with several popular baseline clustering methods.**

I. Introduction

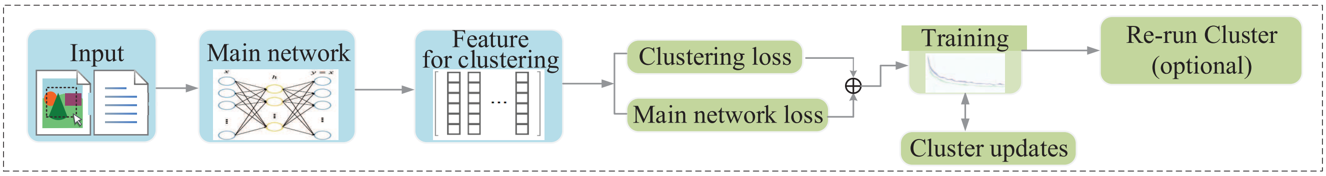
With the development of information collection technology, the number of real-world data accumulation has reached an unestimated level. These massive data are mainly composed of some high-dimensional data, such as text, images, videos, etc. When classifying such a large number of unlabeled data, clustering is an essential data analysis and visualization tools [1]. However, the performance of standard clustering method is adversely affected when coping with high-dimensional data [2] [3]. Thus, tackling the curse of dimensionality, deep clustering (DC) [4] [5], a deep neural networks (DNNs)-based clustering method has become one of the effective ways to solve high-dimensional data categorization. DC is a specific clustering method that combines deep feature representation and clustering algorithm, which utilizes the excellent learning ability of deep learning to mitigate the curse of dimensionality and complete clustering tasks. In this paper, we aim to develop new deep learning solutions to the unsupervised clustering problem.

In most successful methods for clustering with deep neural networks all work following the same principle: representation learning using DNNs and using these representations as input for a specific clustering method [6]. The general pipeline of most clustering methods based on deep learning can be shown in Fig. 1(a). A main network is used to transform the inputs into a latent representation for clustering, and clustering loss can achieve clustering for the given features. After iteration training, the network can update feature representation and clustering results.

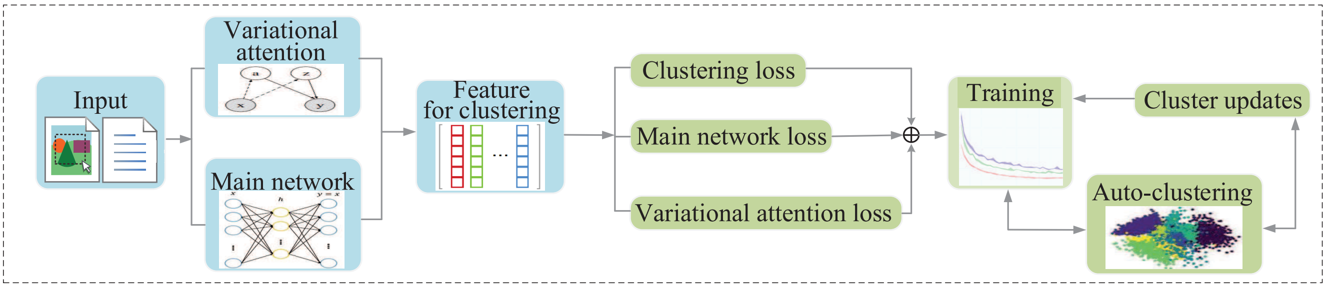
Recently, a lot of promising DC methods have been extensively studied in [7]–[14]. Among those works, deep encoder-decoder framework is one of the popular framework. Specifically, this framework builds on a encoder-decoder framework and via Kullback Leibler (KL) divergence [15] to optimize model. Despite these deep clustering methods have achieved encouraging performance, we argue that they may suffer from some limitations as follows: 1) Just using initial features (z), which are not able to capture the rich features of data; 2) Complicated clustering methods perform poorly owing to unreliable similarity metrics for high-dimensional data, which are not flexible in real-world clustering tasks; 3) Using a two-stage process, namely, first extract feature vectors by DNNs, and then clustering, which leads to a suboptimal clustering result. Next, we will introduce these three limitations in detail.

As the essential problem of clustering task for real-world data, feature representation learning model with neural attention mechanism has been shown effective in various applications. The reason is that the attention mechanism obtains richer implicit feature representation in terms of probability matching. However, the latent feature (z) maybe lose effectiveness when it utilizes traditional attention mechanism directly to process image and text data [16]. It is known as “bypassing” phenomenon [17], which can be shown by a graphical model in Fig. 2(b). This phenomenon can be understood that when hidden layer feature (z) are combined with attention vectors (a), the attention vector (a) will replace hidden feature (z). This will cause the loss of effectiveness of the hidden layer feature (z). In this research, we find that this phenomenon also exists in the auto-encoding framework, resulting in incomplete original information. For this issue, to cope with complex sample distributions and obtain rich feature representation, we utilize variational attention to enhance the feature-aware ability of our model.

There are two types of clustering methods: clustering algorithm-based method and clustering loss-based



(a) The general pipeline of most deep Learning based clustering methods



(b) Our unsupervised deep clustering pipeline based on variational attention and auto-clustering

Fig. 1. The structure of clustering with deep neural networks. Blue is the network block, and green is the optimization block. (a) The general pipeline of most deep-learning-based clustering methods (b) Our variational encoder-decoder feature learning and auto-clustering structure. A framework that incorporates variational attention, automatic clustering mechanisms, and joint optimization of feature learning and clustering.

method [18] [4]. Generally speaking, clustering algorithm-based method has iterative clustering calculation operations. Such a richer feature representation could be favorable for clustering. However, feature representation cannot be updated by feedback from clustering results, and it generates a sub-optimal clustering results due to unreliable similarity metrics for high-dimensional data. Distinct from clustering algorithm-based method, clustering loss-based method is independent of the clustering algorithm and usually enforces a desired constraint on the learned model. The key of clustering loss-based method relies on the construction of clustering constraints and how to joint feature learning and clustering optimization. Based on this, we design a feature-aware automatic clustering module to guide network learning, which can make the network to train in a better direction. In other words, this module avoid the unreliability of sample similarity calculation in clustering task, enabling the more higher performance clustering.

Besides, most existing deep clustering methods are mainly with a two-stage process on the whole, namely, extracting features and clustering. However, the two parts cannot promote each other, and unfriendly feature representation brings difficulties to clustering tasks. Hence, we explore a joint optimization objective approach with feature representation learning and clustering to make mutual promotion possible.

As mentioned before, motivated by the limitation of existing deep clustering, we propose a Deep Variational Attention Encoder-decoder for Clustering (DVAEC). Our work is complementary to existing works in feature representation learning and clustering, since it incorporates the merits of variational attention into deep clustering. The structure and graphical model of our DVAEC are shown in Fig. 1(b) and Fig. 2(c) respectively. As we can see from Fig. 1(b), our DVAEC incorporates variational attention, automatic clustering, and joint optimization of feature learning and clustering. Moreover, it can be seen from Fig. 2(c) that attention feature (a) and hidden layer feature (z) are independent of each other, and it retains the validity of the features of each part. The main contributions of this paper can be summarized as follows.

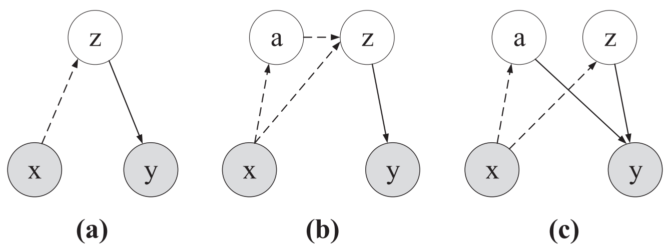


Fig. 2. Graphical model representations. (a) variational auto-encoder (VAE), (b) variational encoder-decoder with traditional attention, (c) variational attention encoder-decoder. Dashed lines and solid lines represent the encoding phase and decoding phase respectively.

- We propose a novel deep variational attention encoder-decoder for clustering that enhances implicit feature representation learning ability. To the best of our knowledge, we are the first to apply variational attention to deep clustering tasks.
- In order to avoid the unreliability of sample similarity calculation in clustering task, we devise a feature-aware automatic clustering module to guide network learning and predict sample category. It can cluster extracted feature directly and eliminate sample similarity calculation.
- To improve the training process of the deep cluster-

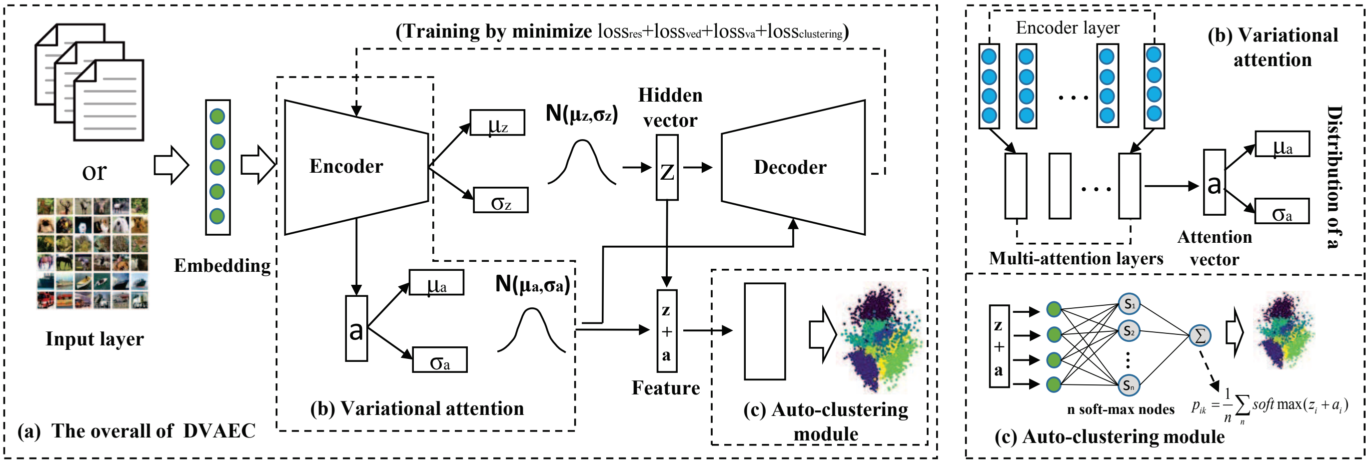


Fig. 3. Illustration of our DVAEC framework, which introduces the variational attention mechanism and auto-clustering module to deep encoder decoder network. The left (a) shows the overall framework of DVAEC. The right (b) shows a detailed variational attention mechanism. The right (c) shows a detailed auto-clustering module.

ing task, we present an optimization objective approach which joints feature representation learning and clustering simultaneously. Extensive experiments on real-world image and text benchmark datasets demonstrate the effectiveness of our DVAEC.

II. Related Work

The proliferation of deep learning has swept the research communtity, among which clustering is no exception. The majority of work that integrates clustering with deep learning methods primarily utilized deep feature representation for clustering task [4] [18] [19] [20].

DEC [4], a structure of simultaneously learns feature representations and cluster assignments using deep neural networks. It does so by letting cluster assignment probability distribution Q approach an auxiliary (target) distribution P which guarantees this constraint. VaDE [8] combines the Variational Auto-encoder (VAE) and Mixture of Gaussian Models (GMM). By calculating the results of distribution, utilizes GMM to pick a cluster. Then it iterates to optimized the evidence lower bound using the stochastic gradient variational bayes estimator and the reparameterization trick. DEPICT [9] proposes a new clustering model which called deep embedded regularized clustering, which efficiently maps data into a discriminative embedding subspace and precisely predicts cluster assignments. IDEC [10] proposes the improvement of importance variational distribution based on DEC. SpectralNet [21] learns a map that embeds input data points into the eigenspace of their associated graph laplacian matrix and subsequently clusters them. In the existing deep clustering methods, attention mechanism is seldom used to enrich the deep representation feature. Variational attention mechanism gives a method in which the original feature and the attention feature are independent of each other, and obtains both features simultaneously. Thus, it is well expected that our DVAEC can obtain satisfactory feature representation in clustering task. Unlike existing deep clustering methods, our DVAEC learns to obtain the low-dimensional feature representation using a variational attention network instead of the previous DNNs.

In general, our work innovates from three perspectives of feature learning, clustering methods and joint optimization. As for feature representation learning, we utilize variational attention to alleviate the "bypassing" phenomenon. In clustering method aspect, we make cluster-friendly feature representation as input for clustering, and construct a clustering function. The task of clustering for high-dimensional data is completed by jointly optimizing feature learning and clustering.

III. Our DVAEC

In this section, we first describe of the clustering problem, and then we introduce the overall architecture of DVAEC, at last we report the optimization objective of DVAEC.

A. Clustering problem definition

We consider that the clustering task denotes to divide N samples into K clusters. X=[x_1, x_2, ... , x_n] $\in R^{d\times n}$, d is the dimension of samples and n is the number of samples. We aim to utilize a deep variational encoder-decoder to obtain the representation of each sample x, and via an auto-clustering mechanism to prediction sample's category.

B. The Framework of Our DVAEC

The overall architecture of our DVAEC is a deep encoder-decoder framework that merges a variational attention mechanism and an auto-clustering mechanism. The framework of DVAEC is shown in Fig. 3.

Input layer: It is to reshape the data as a input of the model. We set different parameters for text and images respectively in terms of data features. For image, we set shape according to pixel dimensions. For text, we first align the length for all samples, and then set the longest dimension as the shape of the input layer.

Encoder layer: Encoding layer consists of multi-convolutional layers(CNN). For the CNN-layer as $Conv_n^k$, n is the number of filter and k is the size of the kernel, the stride as 3. We set $Conv_{64}^3$-$Conv_{64}^3$-$Conv_{128}^5$-$Conv_{128}^5$-$Conv_{128}^5$ as the encoder layers.

Calculation layer: It calculates the mean and variance of the encoder's feature. We propose two posterior distributions for encoder, the recognition model p(z|x)∼N(μ_z,σ_z) as a probabilistic encoder and p(a|x)∼N(μ_a,σ_a) as a attention probabilistic encoder from attention vector. Hence, p(z|x) and p(a|x) can be rewritten as p(z|w,μ_z,σ_z), p(a|x, μ_a,σ_a).

$$p(z|x) \sim \mathrm{N}(\mu_z, \sigma_z), p(a|x) \sim \mathrm{N}(\mu_a, \sigma_a) \tag{1}$$

where x is the input vector, z is the hidden vector, a is attention vector, μ_z,σ_z are mean and variance of z, μ_a,σ_a are mean and variance of a.

Variational attention layer: This layer calculates the attention vector between each embedded encoder layer, and then calculates the mean and variance of the attention vector. The structure of variational attention layer is as shown in Fig.2(b). The attention vector a_{lj} is obtained by a soft-max function from the encoder layer's weight vector e_{lj}. p(a|x, μ_a,σ_a) is sampling from a_i.

$$e_{lj} = Leaky\mathrm{Re}LU(a^T(z(l)|z(j))), (z(l) = Wh_l) \tag{2}$$

e_{lj} is the weight vector between l-th and j-th layer of encoder by LeakyReLU function, | is parameter of connecting the two layers, and $z(l)$ and $z(j)$ is the l-th and j-th encoder hidden vector.

$$a_{lj} = \frac{\exp(e_{lj})}{\sum_{m \in N(l)} \exp(e_{im})}, (l \leq j \in m) \tag{3}$$

$$a_i = \sum_{j \in m} a_{lj}, (l \leq j \in m) \tag{4}$$

where a_i is accumulate by a_{lj}, and $l \leq j \in m$. m is the number of encoder layers.

We define the loss of variational attention in Eq.(5). $loss_{va}$ is to minimize the distribution of attention vector that decoder and encoder. We use the KL divergence [15] to calculate the difference between the two distributions. The effect of $loss_{va}$ is to reconstruct the sample's attention distribution and overcome the "bypassing" phenomenon. The q(x'|a) is a posterior of a.

$$loss_{va} = KL(q(x'|a)||p(a|x)) \tag{5}$$

Hidden layer: The hidden layer is the hidden space (z) of the model. It connects the calculation layer and decoder layer, which is the feature space and as the input of decoder phase. Hidden layer features and attention features are independent of each other. In this way, the two features are guaranteed to be effective at the same time, and preventing one side from being swallowed by the other. When connecting clustering, the two are added as cluster-friendly feature input.

Decoder layer: The decoder layer is the corresponding encoder layer, mainly to reconstruct the input according to the deconvolutional networks. We set $de\!-\!Conv_{128}^5$-$de\!-\!Conv_{128}^5$-$de\!-\!Conv_{128}^5$-$de\!-\!Conv_{64}^3$-$de\!-\!Conv_{64}^3$ as the decoder layers.

We model the posterior q(x'|z)and q(x'|a) as the general model by (z) and (a). Both q(x'|z)and q(x'|a) are the normal distribution defined in Eq.(6). The loss of variational encoder-decoder is defined in Eq.(7). $loss_{ved}$ is to minimize the distribution of samples between decoder and encoder. The effect of $loss_{ved}$ is to obtain hidden features (z) that can be reconstructed by decoding the encoder data.

$$q(x'|z) \sim \mathrm{N}(0,1), q(x'|a) \sim \mathrm{N}(0,1) \tag{6}$$

$$loss_{ved} = KL(q(x'|z)||p(z|x)) \tag{7}$$

Output layer: The output layer of encoder-decoder framework is to reconstruct the input of the encoder. By minimizing the difference between output x' and the input x to train the generation of the model.

The definition of the reconstruct loss is defined in Eq.(8). $loss_{res}$ is a basic loss in encoder-decoder framework, and it drives model to training.

$$loss_{res} = j_{res}(w, \phi, x') = ||x_i{}' - x_i||^{\frac{1}{2}} \tag{8}$$

where x' is the output of the decoder, x is the input data. The reconstruct loss is the fundamental loss of the encoder-decoder framework.

Feature-aware auto-clustering module: We design a feature-aware auto-clustering module to learn similarity calculation and predict category directly instead of traditional sample similarity calculation process. Specifically, we devise a cluster prediction mechanism p_{ik} as a soft-max layer, which consists of n soft-max nodes. We develop the hidden space and attention vector (z+a) as input of the cluster prediction layer to predict classes. The auto-clustering layer obtains more accurate prediction results by weighted average of n soft-max nodes. The auto-clustering layer is defined in Eq.(9). In addition, we define the construct the feedback mechanism q_{ik} in Eq.(12).

$$p_{ik} = \frac{1}{n} \sum_{n} soft\max(z_i + a_i) \tag{9}$$

$$= \frac{1}{n} \sum_{h=1}^{n} P(y_i = k|(z_i + a_i), \theta_h) \tag{10}$$

$$= \frac{1}{n} \sum_{h=1}^{n} \frac{\exp(\theta_k^T(z_i + a_i))}{\sum\limits_{k'=1}^{k} \exp(\theta_{k'}^T(z_i + a_i))} \tag{11}$$

$$q_{ik} = \frac{p_{ik}{}^2/(\sum\limits_{i'} p_{i'k})^{\frac{1}{2}}}{\sum\limits_{k'} p_{ik'}{}^2/(\sum\limits_{i'} p_{i'k'})^{\frac{1}{2}}} \tag{12}$$

where P(y_i=k|z_i) is the probabilistic of sample x_i belonging to the k-th cluster. i' and k' are subsamples and cluster of subsamples. θ=[1,2,..,k]$\in R^{dz \times k}$. θ is the parameter of soft-max f_θ : $(z_i + a_i) \longrightarrow$ k. z_i and a_i are the hidden vector and attention vector of sample i. Clustering prediction task is defined in Eq.(9), and we derive it to get Eq.(11). In order to construct a cluster constraint term, we define q_{ik} in Eq.(12). The function of q_{ik} is to calculate the predicted distribution of sample categories, and combine p_{ik} to construct clustering loss.

The loss of feature-aware automatic clustering is defined in Eq.(13). The effect of $loss_{clustering}$ has two parts. The first is to predict the sample category, and the second is to balance the results of sample prediction more smooth.

$$loss_{clustering} = KL(q_{ik}||p_{ik}) \tag{13}$$

C. Optimization of DVAEC

The overall training objective of DVAEC consists of four parts: $loss_{ved} + loss_{va} + loss_{ckustering} + loss_{res}$. We show the objective is to minimize in Eq.(14).

$$\begin{aligned} &J(w,\phi) = j_{res}(w,\phi,x') + \lambda KL(q(x'|z)||p(z|x)) \\ &+ (1-\lambda)\, KL(q(x'|a)||p(a|x)) + KL(q_{ik}||p_{ik}) \end{aligned} \tag{14}$$

Here, j_{res} is the reconstruction loss of the encoder-decoder, w and ϕ are the parameters of the encoder-decoder layer. KL(q(x'|z)||p(z|x) [15] is the distribution loss between encoder and decoder. KL(q(x'|a)||p(a|x) is the variational attention distribution loss between encoder and decoder. Especially, $\lambda \in [0, 1]$ is a hyper-parameter which is used to balance variational attention loss and variational encoder-decoder loss. In the section of experiments, we will introduce the effect of the parameter λ on the performance of the model. KL(q_{ik})||p_{ik})) is the cluster constraint which aims to the clustering effect smooth. We use Adam to optimize the objective in Eq. (14).

The lower bound of DVAEC. Since the likelihood of the DVAEC aggregates three KL divergence, we propose the variational lower bound of DVAEC in Eq. (15).

$$\begin{aligned} &\mathcal{L}_i{}^{(n)}(w,\phi) = \mathbb{E}_{z,a\sim q(z,a|x^{(n)})}[\log p(x'^{(n)}|z,a)] \\ &-\lambda KL(q(x'|z)||p(z|x)) - KL(q_{ik}||p_{ik}) \\ &-(1-\lambda)\, KL(q(x'|a)||p(a|x)) \end{aligned} \tag{15}$$

In Eq.(15), the variational attention (a) and the latent feature space (z) are independent in recognition and reconstruction. The posterior factorizes as q(x'|z)and q(x'|a), because (z) and (a) are independent given x (dashed lines in Fig. 2(c)). Whereas the prior factorizes (z) and (a) are marginally independent (solid lines in Fig. 2(c)). In this way, the sampling procedure can be done separately and the KL loss can also be computed independently [17]. Besides an auto-clustering constraint KL($q_{ik}||p_{ik}$) is independent, and it can be calculated directly.

IV. EXPERIMENTS

To verify the effectiveness of the our DVAEC, in this section, we first introduce the experiments settings, and then analyze the experimental results compared with several popular methods.

A. Experiments settings

*1) **Data description and Evaluation Metrics**:* We evaluate the performance of DVAEC on 6 real-world data-sets which includes 3 image datasets and 3 text datasets. The detail of data is shown in Table 1.

Fashion-MNIST [22][1]: a dataset consists of 60,000 images labeled as 10 classes.

CIFAR-10[2]: a dataset consists of 60,000 images labeled as 10 classes.

USPS[3]: a dataset contains 9298 grayscale images.

20NEWS[4]: a popular database for text classification or clustering. We use 4 categories.

REUTERS[5]: a dataset has 810,000 English news, following DEC [4].

StackOverflow[6]: a collection of posts from question and answer site stackoverflow, published as part of a Kaggle challenge [21].

TABLE I
DETAILED DESCRIPTION OF THE DATA SETS. NUMBER OF SAMPLES (N), NUMBER OF CLASSES (K), SIZE OF IMAGE DATE, NUMBER OF VOCABULARY (V) FOR TEXT DATA.

Dataset	N	k	size	V.
Fashion-MNIST	60k	10	28×28×1	-
CIFAR-10	60k	10	28×28×3	-
USPS	9.3k	10	16×16×1	-
20news	3k	4	-	2.8k
REUTERS	10k	10	-	13k
StackOverflow	20k	20	-	23k

Evaluation Metrics. We use clustering accuracy **(ACC)** [4] and normalized mutual information **(NMI)** [4] to evaluate the clustering performance of the models.

*2) **Compared Methods**:* To evaluate the performance of DVAEC, we compare it with several popular baseline methods. Especially, to compare the representation learning performance of our models, we set three comparison methods based on K-means. For the accuracy of the overall clustering task, we select 4 popular comparison models.

K-means+AE [4]: It runs k-means algorithm in the AE hidden feature space.

K-means+VAE [15]: It runs k-means algorithm in the VAE hidden feature space.

K-means+Our(z): We run k-means algorithm in our DVAEC single hidden embedding space (z).

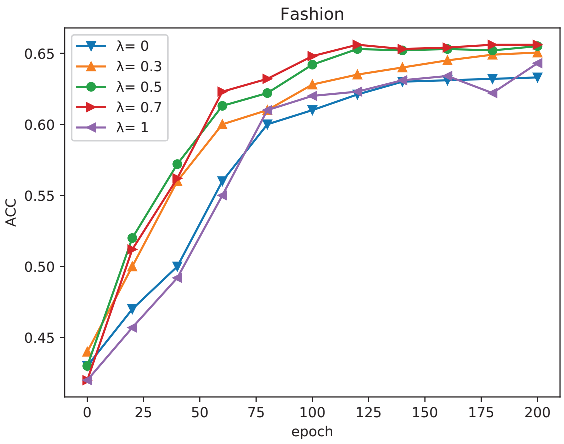
K-means+Our(z+a): We run k-means algorithm in our DVAEC joint feature consist of hidden embedding space(z) and attention vector (a).

DEC [4]: This is the first deep embedding clustering based on SAE and K-means. We set parameters in terms of this model.

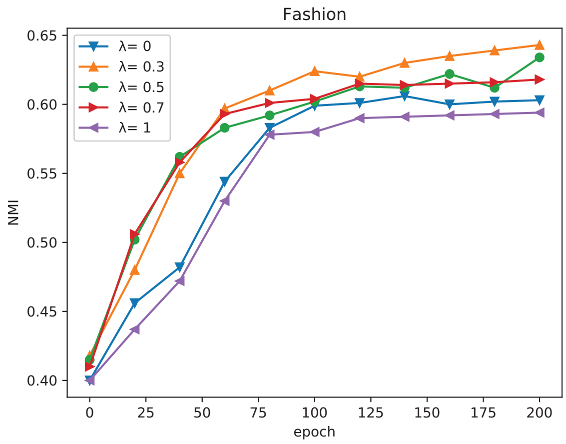
[1] http://fashion-mnist.s3-website.eu-central-1.amazonaws.com/
[2] https://www.cs.toronto.edu/ kriz/cifar.html
[3] https://www.csie.ntu.edu.tw/ cjlin/libsvmtools/datasets/multiclass.html
[4] http://qwone.com/jason/20Newsgroups/
[5] https://github.com/philipperemy/Reuters-full-data-set
[6] https://www.kaggle.com/c/predictclosed-questions-on-stack-overflow/

TABLE II
THE RESULTS OF OUR PROPOSED FRAMEWORK AND SEVERAL STRONG CONTRAST METHODS ON 6 BENCHMARK DATA-SETS IN ACC (%) AND NMI (%). THE BEST RESULTS ARE MARKED WITH BOLD SYMBOL.

Methods \ Datasets	Fashion		CIFAR-10		USPS		20news		REUTERS		StackOverflow	
	ACC	NMI	ACC	NMI	ACC	NMI	ACC	NMI	ACC	NMI	ACC	NMI
K-means+AE	47.5	51.5	21.8	10.1	65.3	62.8	33.76	0.71	53.3	52.7	40.1	39.2
K-means+VAE	50.3	51.4	28.0	23.4	70.3	61.1	45.3	23.1	73.5	70.7	42.7	40.5
K-means+our(z)	52.0	51.6	31.2	32.6	71.8	68.4	51.8	47.1	77.2	74.3	45.3	44,7
K-means+our($z+a$)	53.4	53.6	32.3	31.7	73.3	72.5	51.9	49.0	78.6	76.3	46.1	45.2
DEC [4]	51.6	54.6	26.3	25.7	74.1	74.3	50.1	44.4	75.6	70.4	46.3	45.6
IDEC [10]	52.9	55.7	25.1	24.7	76.2	78.5	53.6	44.5	-	-	47.9	46.2
VaDE [8]	58.2	57.3	44.6	45.2	76.9	71.2	67.4	63.5	79.2	72.7	47.1	46.5
SpectralNet [21]	60.1	58.7	48.3	**46.7**	82.5	**80.4**	73.4	71.5	82.1	80.0	52.7	50.4
DVAEC (ours)	**64.7**	**61.4**	**50.6**	42.6	**84.5**	79.3	**75.1**	**70.0**	**83.5**	**81.7**	**56.4**	**53.7**



(a)



(b)

Fig. 4. Impact of the parameter λ on the Fashion-mnist dataset (image). It shows that the variation of ACC (a) and NMI (b) with epoch sizes. The main contrast is the influence of parameter λ on our model when processing image data. It is limited that only lambda changes, and the remaining parameter settings are consistent.

IDEC [10]: It is an improved version of DEC with local structure preservation.

VaDE [8]: VaDE is a framework of clustering combined with representations, it utilizes the Gaussian mixture model(GMM) to optimize the variational auto-encoder.

SpectralNet [21]: SpectralNet is a clustering approach combined with spectral clustering and deep network.

B. Clustering performance evaluation

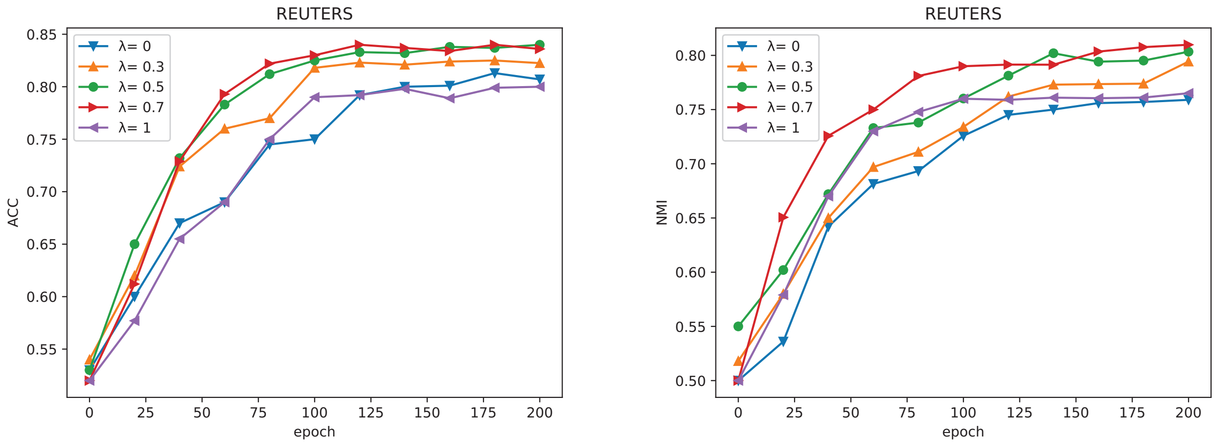
We evaluate the performance of the model from the aspect of feature learning and the overall performance of clustering. The clustering results are shown in Table 2.

First, we compare four methods for feature representation, and there is K-means clustering algorithm for clustering 4 features (hidden feature of AE, VAE, the hidden feature of DVAEC(z) and hidden of DVAEC add attention vector(z+a)). The feature of AE and VAE, we utilize the same number of network layers as our network. We employ basic auto-encoder(AE) and variational automatic encoder(VAE) frameworks which the same as [4]. As shown in Table 2, compared with the hidden layer features of the AE and VAE model, the single hidden feature (z) of DVAEC has improvements of 4.5% and 1.7% in metric of ACC on fashion dataset. When merging the attention feature (z + a), the accuracy improves average 5.8% and 3.9% compared with AE and VAE on fashion dataset. It shows the better the feature, the more accurate the result of clustering calculation. By limiting the conditions of the same clustering method, the experimental results show the representation learning ability of our DVAEC.

Besides, we compare with 4 baselines models on the overall performance of the model (DEC, IDEC, VaDE, SpectralNet). Compared with DEC and IDEC, our DVAEC clustering performance improves 10.1% and 8.5% in terms of ACC on StackOverflow dataset. DVAEC adopts a variational attention mechanism, and it utilizes (a+z) as the input of clustering layer. In addition, the cluster prediction layer picks the category to avoid the similarity calculation of samples. DVAEC also gets 9.3% and 3.7% improvements over VaDE and SpectralNet on StackOverflow dataset. It shows the high clustering performance of our DVAEC.

C. Model and Parameter analysis

In this section, we aim to analyze the structure of the DVAEC model. The overall framework of DVAEC is shown



(a)

(b)

Fig. 5. Impact of the parameter λ on the REUTERS dataset (text). It shows that the variation of ACC (a) and NMI (b) with epoch sizes. The main contrast is the influence of parameter λ on our model when processing text data. It is limited that only lambda changes, and the remaining parameter settings are consistent.

in Fig. 3, and graphical model representation is shown in Fig. 2(c).

The effect of variational attention and auto-clustering. It can be clearly seen in the structure of Fig. 2(c). Our DVAEC based on variational attention mechanism is independent of the variational encoder-decoder structure to solve the "bypassing" phenomenon. By constructing a variational attention loss, the learning ability of the model is further improved without affecting the clean of the feature (z). Besides, we devise a novel feature-aware auto-clustering module to avoid the calculation of sample similarity matrix. In addition, the more high-quality mix-feature (z+a) improves the accuracy of cluster prediction.

Parameter selection. In our DVAEC, we design the structure of encoder of embedding layers as a multi-layer convolutional neural network for all sets. The decoder embedding layer is de-CNN networks. We set dropout = 0.5, mini-batch = 128. In text cluster task, we use the BERT [23] pre-training model to initialize word embedding, d_w=300. The learning rate of the SGVB lr = 0.01. The hidden layer space dimension (z) sets 40. For the auto-clustering mechanism is shown in Eq.9. n is number of the softmax nodes, and we set n = 8.

Especially, we perform a test on the effect of the parameter λ on the metrics of ACC and NMI. As shown in Fig. 4 and Fig. 5. When setting different values of λ , the ACC and NMI changed with the epoch addition. It is clear that whether the dataset is an image or text when the λ is setting between [0.5-0.7], the ACC and NMI are better. Compared with the model without variable attention, the accuracy and NMI are improving of the model when add the variable attention. Therefore, we set λ=0.6 for all experiments.

Discussions. As mentioned above, we design an unsupervised clustering framework that incorporates variational attention and feature-aware auto-clustering. From the experimental results, we can observe that:

(1) Our DVAEC framework achieves the best performance on both datasets, significantly outperforming the state-of-the-art DEC and IDEC methods. On average, the relative improvement over the baseline SpectralNet and VaDE methods is 2.7% and 4.6% respectively. The reasons are that our DVAEC first utilizes variational attention to obtain more richer features and proposes a feature-aware auto-clustering module to achieve more accurate clustering results.

(2) We introduce an adjustable parameter λ to balance the deep representation learning and auto-clustering in our DVAEC framework. When the value of λ is between 0.5 and 0.7, our DVAEC obtains the best performance compared with other baseline methods.

(3) In terms of model optimization, we present an overall optimization objective to train our DVAEC. Specifically, the optimization objective consists of four parts, namely $loss_{ved}+loss_{va}+loss_{ckustering}+loss_{res}$. In this way, we integrate variational attention into deep representation learning. Meanwhile, we aovid the complicated iteration computing and complete the auto-clustering task by updating the results constantly.

V. Conclusions

In this paper, we aim to incorporate variational attention mechanism to enhance representation learning and complete clustering tasks effectively. We introduce variational attention mechanism to enhance the feature representation for deep clustering. Besides, we design a novel feature-aware auto-clustering module to predict the cluster of the sample precisely. Finally, we propose a joint optimization object for the sake of better jointly optimize feature learning and clustering. Extensive experiments on six real datasets demonstrate that our method outperforms several popular models.

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